[6]

Hosting Services and Authentication

In the last chapter, you learned about merging and how this feature of Git allows you to integrate changes from one branch into another.
Up until this point we have discussed working only in local repositories, and you have worked only on the rainbow repository, which is a local repository.
This chapter marks the start of the second part of this book, in which you will work with hosting services and remote repositories.
In this chapter, you will choose a hosting service and prepare the authentication details you will use to connect to remote repositories on that hosting service by using either the HTTPS (Hypertext Transfer Protocol Secure) or SSH (Secure Shell) protocol. The information that you will need to carry out these tasks is contained in this chapter as well as in the resources in the Learning Git repository (*https://github.com/gitlearningjourney/learning-git*).

[NOTE]

If you work at a company that uses Git and a hosting service, you may already have a hosting service account using your company email address. However, I recommend using a personal account with a hosting service for the exercises in this book instead of a company account. This is because your company may have configured additional settings on the company hosting service account that could complicate exercises later in the book.

If you already work with a hosting service that you want to use to complete the exercises in this book and you already have authentication details set up to connect to remote repositories over a secure protocol, then you may skip this chapter and move on to Chapter 7 to learn more about how to create and push to remote repositories.

If you haven't yet chosen which hosting service you want to use, have not set up authentication details for a specific protocol, or want to learn more about the available protocols, then keep reading.

Hosting Services and Remote Repositories

In Chapter 2 I mentioned two types of repositories: local repositories and remote repositories. Local repositories are found on a computer, while remote repositories are hosted on a hosting service in the cloud.

I also mentioned that hosting services are companies that provide hosting for projects using Git. In this book, I will provide information about the three main Git hosting services: GitHub, GitLab, and Bitbucket.

To transfer data between a local repository and a remote repository on a hosting service, you must connect and authenticate using either SSH or HTTPS. From Chapter 7 onward, you will be uploading and downloading data to and from a remote repository using commands such as git push, git clone, git fetch, and git pull. To use these commands and connect to the remote repository you will have to have prepared authentication details for the protocol you want to use beforehand.

In this chapter, first you will choose a hosting service, and then you will set up authentication details to connect to remote repositories over HTTPS or SSH.

Setting Up a Hosting Service Account

As mentioned previously, if you already work with a hosting service and know you want to use that one for the exercises in this book, you can simply log in to your account on the hosting service website.

If you've never worked with a hosting service, you will have to choose which one you want to use and create an account. The three main hosting services are GitHub, GitLab, and Bitbucket. If you're not sure which one to use, then I recommend using GitHub since it is the most popular one and the one I use for the example output for the Rainbow project in this book.

Go to Follow Along 6-1 to make your choice of hosting service and log in.

[FOLLOW ALONG 6-1]

1. Choose the hosting service on which you want to create a remote repository, and go to the website. Either log in or create an account. In these examples, we will use GitHub.

Now that you have chosen and logged in to your hosting service, you must choose which protocol you want to use to connect to remote repositories and prepare your authentication details for that protocol.

Setting Up Authentication Credentials

When you create a remote repository, there will be two ways to make changes to it:

1. By logging in to the hosting service via its website and making changes there directly.
2. By making changes in your local repository and uploading those changes to the remote repository on the hosting service.

In both cases, you will need to *authenticate*, or verify your identity. Authentication is important, to control who logs in to your hosting service account and who uploads changes to a remote repository.

In the first case, you will use your username or email address and password to authenticate and log in to your hosting service account.

But what about in the second case? How will your hosting service know that it should allow you to upload things from a local repository to a remote repository?

The answer is that you will have to authenticate through the protocol you choose to use. In this chapter, we will cover the HTTPS and SSH protocols. You need to set up only one of these, and you can choose which one you prefer.

[NOTE]

In Chapter 7 you will go through the process to create a remote repository for your Rainbow project, which will produce two URLs: an HTTPS URL and an SSH URL. You will have to use the URL for the protocol you decide to use.

We'll cover the HTTPS protocol first. If you only want to learn about and use the SSH protocol, then you may skip the following section and go straight to "Using SSH" on page 93. Keep in mind that you don't need to understand these protocols in depth in order to use them.

If you're not sure which protocol to choose, I recommend setting up the HTTPS protocol to carry out the exercises in this book because the process is slightly simpler and the examples in this book will use the HTTPS protocol. Whichever protocol you choose to set up, you can always choose to set up the other one as well in your future work with Git, so this isn't a critical decision right now.

USING HTTPS

The HTTPS protocol uses a username and some sort of password (or authentication credential) to allow you to securely connect to remote repositories. In the past, all the hosting services allowed you to use the password you use to log in to your account on the hosting service (which we will refer to as the *account password*) for HTTPS authentication as well. However, GitHub and Bitbucket no longer allow this; they require you to create another authentication credential.

In GitHub, the authentication credential is called a *personal access token*. In Bitbucket, it's called an *app password*. With GitLab, you can still simply use your account password to authenticate. See Table 6-1 for an overview of the three most common hosting services and the access credentials necessary to authenticate over HTTPS.

TABLE 6-1. Hosting services and necessary access credentials for authenticating over HTTPS

HOSTING SERVICE	USERNAME	PASSWORD
GitHub	Email address or username	Personal access token

HOSTING SERVICE	USERNAME	PASSWORD
GitLab	Email address or username	Account password
Bitbucket	Email address or username	App password

Go to Follow Along 6-2 to prepare your authentication credential.

[FOLLOW ALONG 6-2]

1. If you are using GitHub or Bitbucket and you have chosen to use the HTTPS protocol, then go to the Learning Git repository (*https://github.com/gitlearningjourney/learning-git*) for more information on how to set up your authentication credential. You may skip the following section.
   If you have decided to use the SSH protocol, continue on to "Using SSH."

USING SSH

The SSH protocol uses a public and private SSH key pair to allow you to securely connect to remote repositories. The three main steps to setting up SSH access are:

1. Create an SSH key pair on your computer.
2. Add the private SSH key to the SSH agent.
3. Add the public SSH key to the hosting service account.

Go to Follow Along 6-3 to carry out the three steps to set up SSH access.

[FOLLOW ALONG 6-3]

1. If you have chosen to use the SSH protocol, then go to the Learning Git repository (*https://github.com/gitlearningjourney/learning-git*) for more information on how to set up authentication details to connect over SSH.

[NOTE]

Sharing your private SSH key is a security risk. Think of it as a password, and do not share it with anyone.

Now that you have prepared your authentication details to connect to remote repositories over either HTTPS or SSH, you may continue to the next chapter to create the first remote repository for the Rainbow project.

Summary

In this chapter, you chose which hosting service you wanted to use to complete the rest of the exercises in this book, and you prepared the authentication details you will need to connect to remote repositories over either HTTPS or SSH. In Chapter 7, you will learn about creating remote repositories and start exploring how you can work with others on Git projects.

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Chapter 7. Creating and Pushing to a Remote Repository ›